ATTU MK II

USER MANUAL 1.7

(FIRMWARE 1.5.0)

Last update: 15/01/2026
Version: 1.7

Contents

1 Introduction 3

2 Specifications 4

3 ATTU casing and connections 5

3.1 RJ45 serial connector pinout 5

4 Connecting for the first time 8

5 Configuration 9

5.1 Time 9

5.2 Ports 12

5.3 Broadcast Destinations 15

5.4 System information 16

6 Integration 17

6.1 Connecting to NaviPac 17

6.2 Connecting to NaviScan 19

6.3 Combined NaviScan and NaviPac 22

7 Troubleshooting 24

7.1 Occasional yellow flags are showing on the incoming port 24

7.2 Data doesn't fit into the entire box in the incoming port 24

7.3 Unable to connect to ATTU 25

7.4 Time telegram not being received and interpreted 25

7.5 Data not consolidated 26

7.6 The ATTU does not respond 26

7.7 The ATTU is not showing all the data 26

1 Introduction

A survey setup is often a hybrid solution of serial data input from navigation and survey sensors to acquisition software where time of reception of data is very important for the migrated result. A typical survey computer is not the optimal solution for handling time tagging of such data since it needs to handle not only the data presentation and logging, but also user-friendly computer OS.

This is where the ATTU comes in handy, as it is running with a stripped-down LINUX kernel with its only purpose of time tagging serial RS232 sensor input data and is therefore considerably faster and more accurate than a standard personal computer. The sensor data can then be broadcast on network allowing computers running EIVA NaviSuite acquisition software to receive the data and not worry about potential data transmit latency.

The ATTU is used exclusively by the EIVA NaviSuite acquisition software packages only. It is not compatible with any other software packages. It is currently not possible to retransmit the PPS signal to any other device.

This document is written for the ATTU Mk II with firmware version 1.5.0.

2 Specifications

Description	Requirement	Notes
Supply Voltage	100 to 250 VAC	PF ≥0.9, 80Vmax
Frequency	47 to 63 Hz	
Current	1A @ 115V, 0.5A @ 230V	At max power
Power	Typical 35W, max 75W	
Input connector	IEC 60320 - C14	Use IEC C13 cable
Fuse	T3.15A HRC	5*20mm HRC Requires upstream protection 10A max
EMC	Meets IEC 60945	
Weight	5.1 kgs	
Dimensions	483 x 44 x 332 mm (w x h x d)	1U 19 Rack mount
Working Temperature (enclosure):	-30 to 65 °C	
Storage Temperature:	-40 to 90 °C	
Relative humidity:	20 to 95 % RH (non-condensing)	`
Accuracy	< 50 µSec	
Interfaces to sensors	RS232	
Ports	24	
Independent of sensor data format	Yes	
Network time server (NTP) for other units in the network	Yes	
Time synchronisation	Via GPS or NTP	

Table 1 Specifications

3 ATTU casing and connections

The front side has the power button, a green LED which flashes accordingly if the ATTU receives PPS (and time input) from GNSS receiver – typically more accurate than an alternative input from NTP server – and a USB port, which is only used for service.

Figure 1 Front of ATTU unit with power button, LED inidicator, and USB port for servicing

The back side has 24 RJ45 ports for serial data input, and 2 Ethernet ports for monitoring/configuration and broadcast of data.

Each serial port can also send out an initialisation string (one time at startup) for the serial sensors connected – though this is not commonly used.

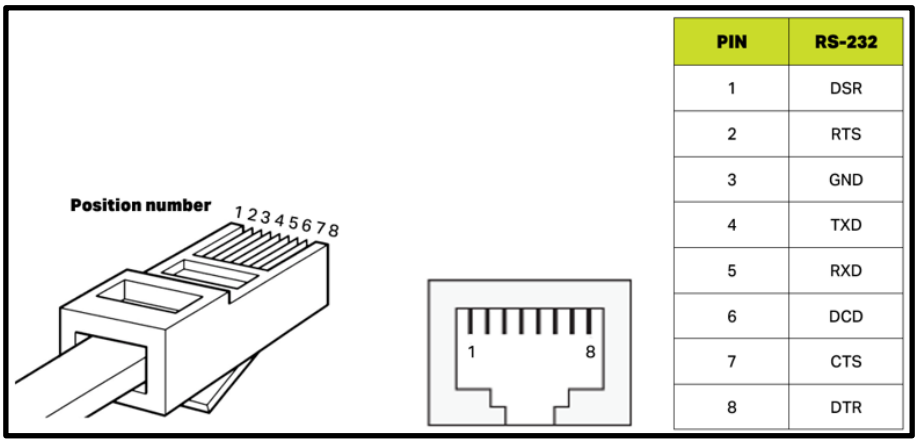
There is also an IEC 14 socket with mains fuse and power on/off switch.

Figure 2 Back of ATTU unit with RJ45 data input ports, ethernet output ports, power input and switch

3.1 RJ45 serial connector pinout

The serial RJ45 connectors pinout are shown in Figure 3 below. The PPS – including time telegram – can be input (must be combined in one input port in RS232 format) to any of the 24 ports for the internal time sync of the ATTU, unless ATTU is timed by NTP.

The PPS signal must be wired to pin 6 (DCD) and ground part of the signal to pin 3 (GND). The time telegram (ZDA or UTC) and other serial input data must be wired to pin 5 (Rx) and 3 (GND).

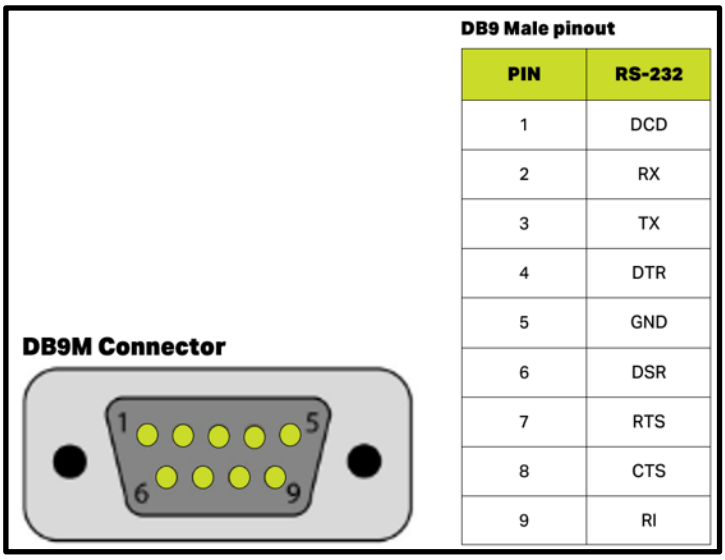


PIN	RS-232
1	DSR
2	RTS
3	GND
4	TXD
5	RXD
6	DCD
7	CTS
8	DTR

Figure 3 Serial connector and pinout

Moxa RJ45–DB9M (male) adaptors (Moxa p/n CBL-RJ45M9-150) are available with the below DB9M pinout. PPS must then be wired to pin 1 (DCD), TX (from sensor) wired to pin 2 (RX) and GND to pin 5.

Caution: Similar RJ45–DB9M adaptors exist with different pinouts and will not match!



DB9 Male pinout

PIN	RS-232
1	DCD
2	RX
3	TX
4	DTR
5	GND
6	DSR
7	RTS
8	CTS
9	RI

Figure 4 DB9M male adaptor and pinout

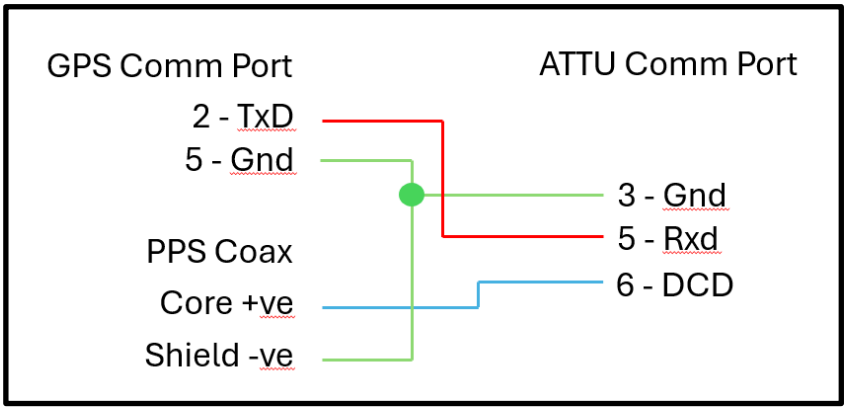


Figure 5 PPS and ZDA connections

Figure 5 shows an example of wiring for the ZDA and PPS pulse into one of the ports. Any port can be defined as the TIME SYNC port.

Some GPS units have the PPS pulse included in the serial data connection and can be wired directly to the DCD pin of the ATTU port. Figure 5 shows an example of a GPS unit that has a separate PPS connection and a ZDA string combining into the Time Sync port of the ATTU.

4 Connecting for the first time

The ATTU IP addresses are shown on the **System Information** page. By default, Ethernet 1 is a static address of 10.10.114.114, netmask 255.255.0.0 and gateway 10.10.0.1. Ethernet 2 is DHCP enabled from factory but can be configured for a static address.

Connect a LAN cable to Ethernet 1 and make sure the computer has a fixed IP address configured within the same IP scope (eg 10.10.114.100). Open an Internet browser and type in the IP address of the ATTU.

Note: Please make sure to note down the IP address of the ethernet IP addresses once configured.

Important: Ethernet 2 will only be displayed in the web interface if there is an active network cable connected.

Both network addresses can be used to show the ATTU status in the browser.

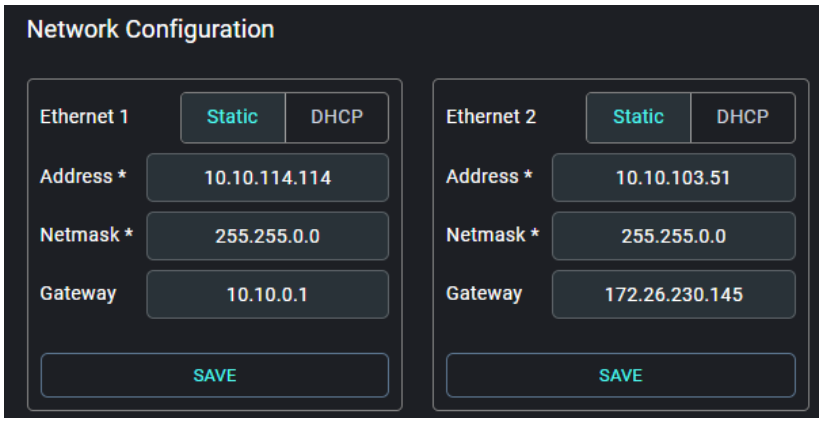


Figure 6 Network Configuration

5 Configuration

5.1 Time

The front page shows the current time and date, the timing input settings and a scope to display the PPS and ZDA inputs.

Figure 7 shows the **Time** page where the time source is configured, either by a GPS PPS pulse and time ZDA datagram (most accurate), from an NTP server (either Internet-based or configured on the local network) or the internal RTC time clock.

Figure 7 Time

Figure 8 shows the ATTU is configured for PPS input with a rising edge, PPS + Datagram (usually PPS+ZDA) on serial port 9. It is possible to have a reversed input pulse order (typical Trimble UTC) Datagram + PPS.

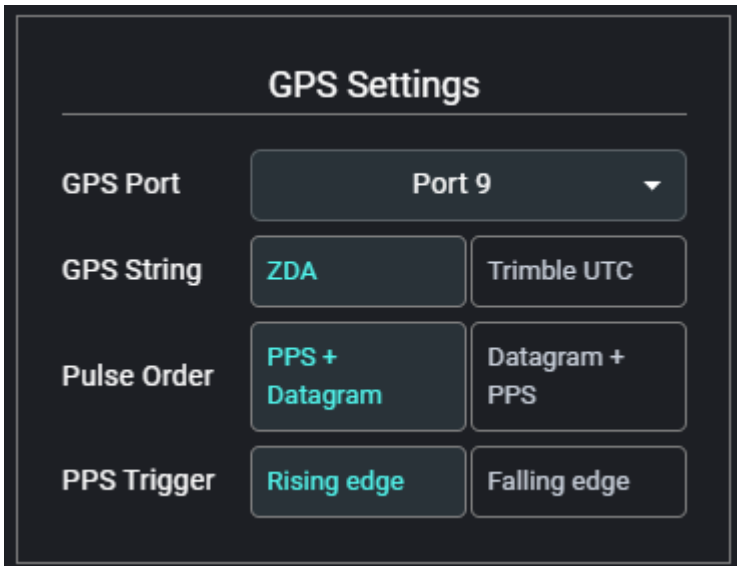


Figure 8 Pulse order and GPS string

If PPS and ZDA are present, they will show up in the window with the pulses and datagrams received within the last minute. Datagram latency is the time between pulse and datagram in microseconds and shown graphically in the graph between the spike (for the PPS Pulse) and the dash (for the ZDA message).

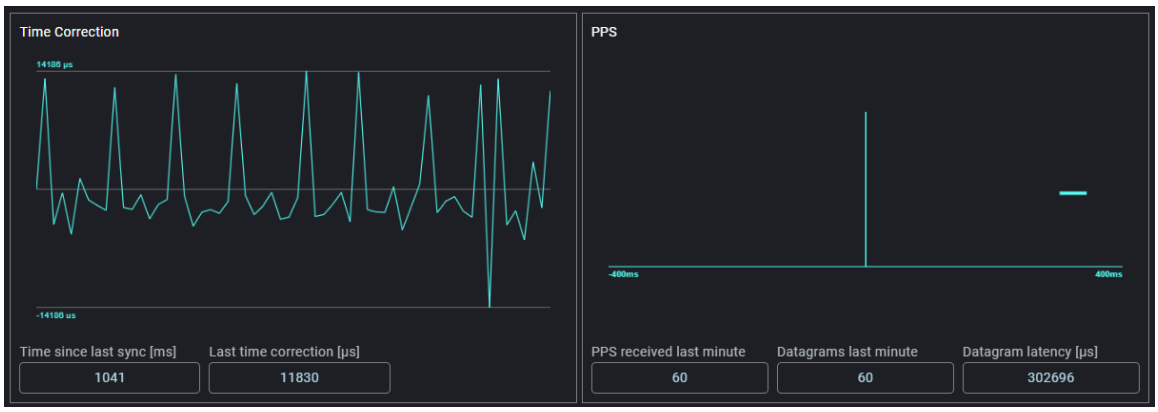


Figure 9 PPS graph

As an alternative and when a GPS unit is not available, a Network Time Protocol (NTP) connection can be used when the ATTU is connected to the Internet or to a local network NTP server.

Any Internet time server such as *time.windows.com* (20.101.57.9), *time.google.com* (216.239.35.0) or any NIST Time Server (129.6.15.28) can be used.

NTP filtering allows for the occasional readings that are out of range from the NTP server to be ignored.

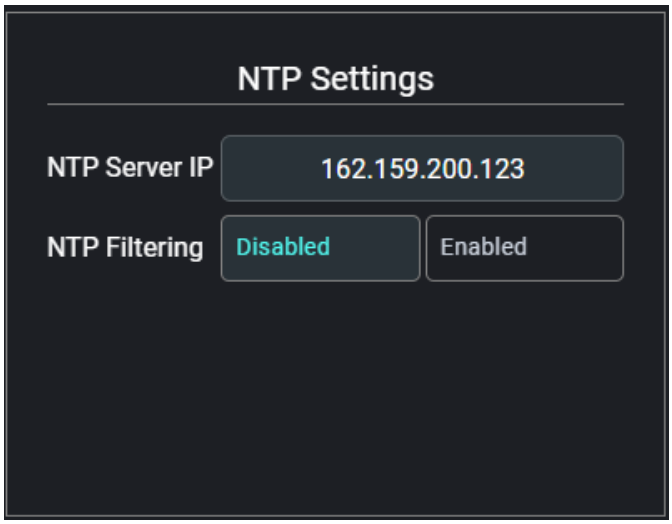


Figure 10 Time and NTP Server

The **Active Time Source** allows the user to select the GPS or NTP (Network Time Protocol).

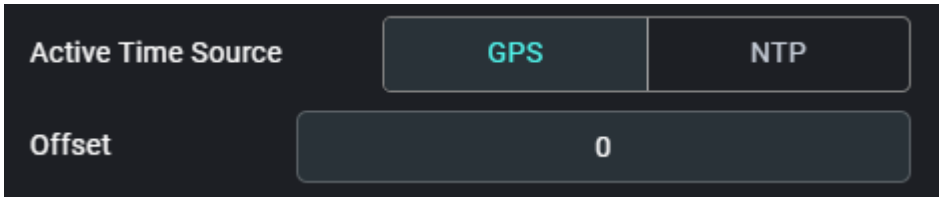


Figure 11 Selecting Time source

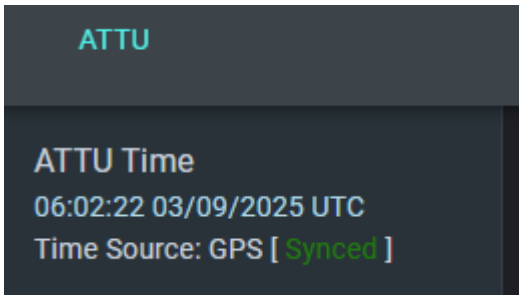


Figure 12 Displaying the source of Time

The messages for the GPS are *Synced* or *Not Synced*. Loss of the PPS pulse or the ZDA message will trigger an error.

The messages for NTP are *Synced* or *Not Synced*. The ATTU will attempt to contact the designated time server for an updated time stamp.
The Real-Time Clock (RTC) uses the ATTU internal clock to keep track of time but will drift and should not be used in place of the GPS or NTP Time Server. It will always say that it is *Not Synced*.

5.2 Ports

The Ports page shows the data being received at each of the 24 ports and whether the data is being interpreted correctly by the ATTU.

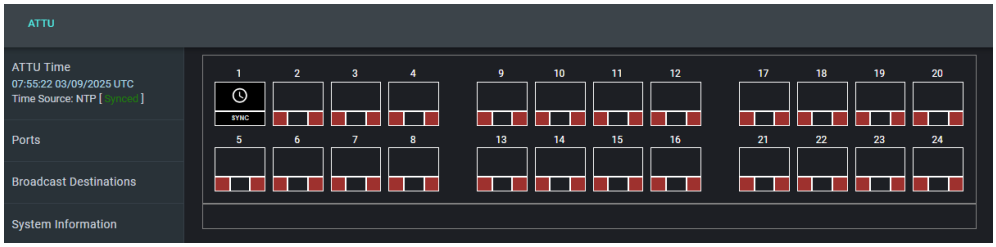


Figure 13 Ports where data is received

Each port is configurable, including labelling each port for ease of identification. After changing the parameters, click **SAVE** to implement the changes. Pressing ESC will close the dialogue box without saving.

Note: Please be warned that clicking outside of the dialogue box will not save the settings. Only by clicking **SAVE** will the settings be changed.

The **SYNC** indicator shows the selected port for the PPS/ZDA input into the ATTU. By default, this port is COM 1 but can be any of the 24 ports.

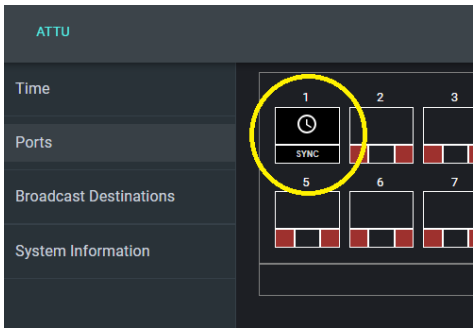


Figure 14 SYNC indicator

Clicking on each port brings up the settings for the port including naming the port for easy identification.

The port choices are industry standard, as well as the option for ASCII and binary.

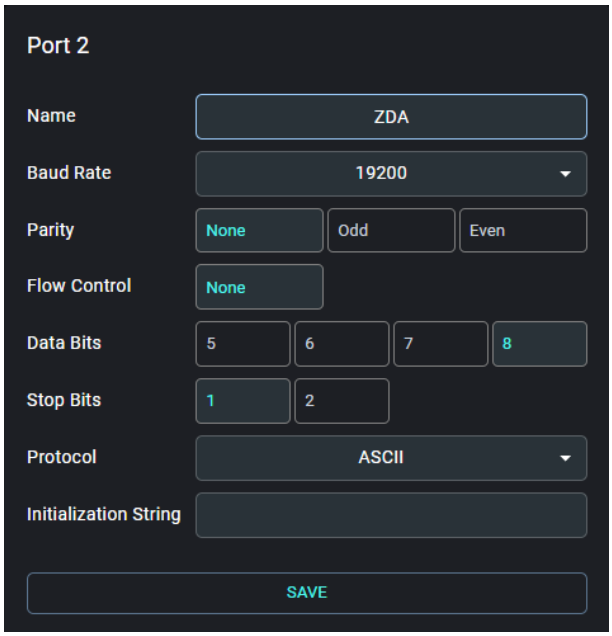


Figure 15 Port settings

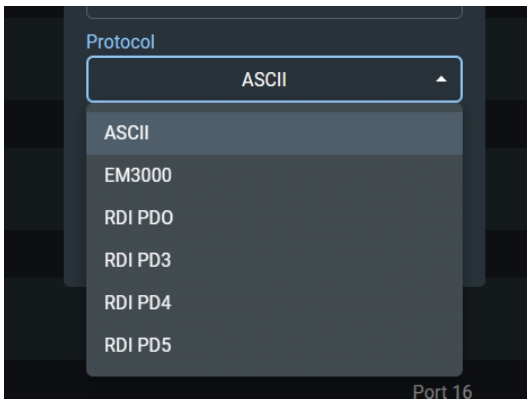


Figure 16 Port choices

Once the data is being received correctly and interpreted by the ATTU, the header of each message will show the data as a string. EM3000 data is in binary and is not readable like the GGA and HDT ASCII data strings.

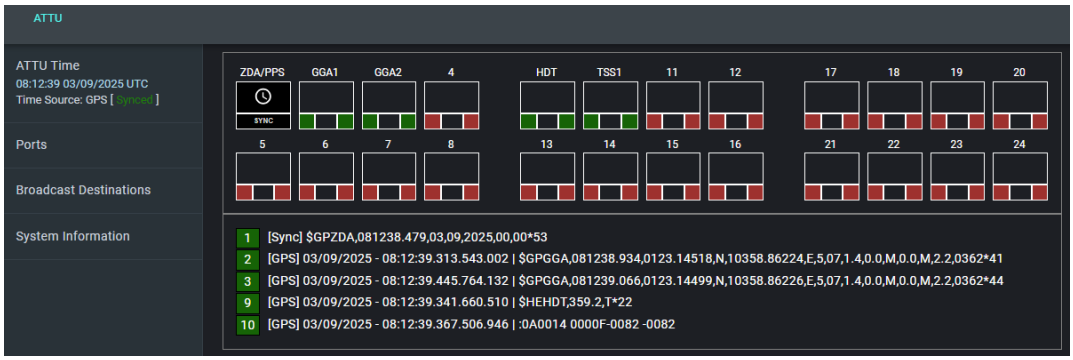


Figure 17 Ports and data being received shown in headers

The port will also display the current source of the time, ie GPS, NTP, as the Internet time or RTC as the internal clock.

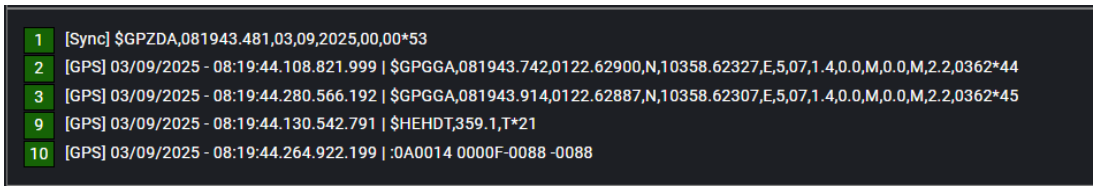


Figure 18 GPS time tag

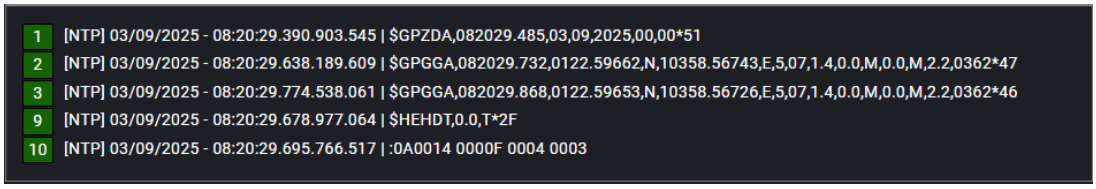


Figure 19 NTP time tag

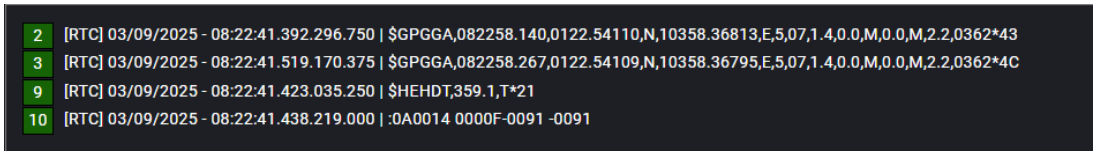


Figure 20 Internal PC clock

5.3 Broadcast Destinations

The **Broadcast Destinations** page shows the details of the destination IP addresses of the receiving computers, and the ports selected once configured.

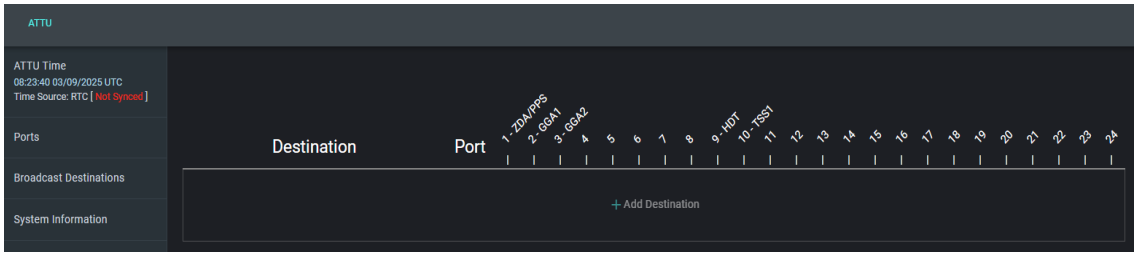


Figure 21 Broadcast page when first shown

The view after adding the destination of the NaviPac and NaviScan computers is shown below, with the allotted ports for the required messages.

Figure 22 Destination of NaviPac and NaviScan computer with allotted ports

Integration into NaviPac and NaviScan will be covered in Chapter 6 Integration.

5.4 System information

The **System Information** page shows the current firmware version as well as the IP addresses of the ethernet ports. Ethernet 2 settings will show up once connected to an active ethernet port and is, by default, set to receive IP address from DHCP server. Click **SAVE** to activate settings.

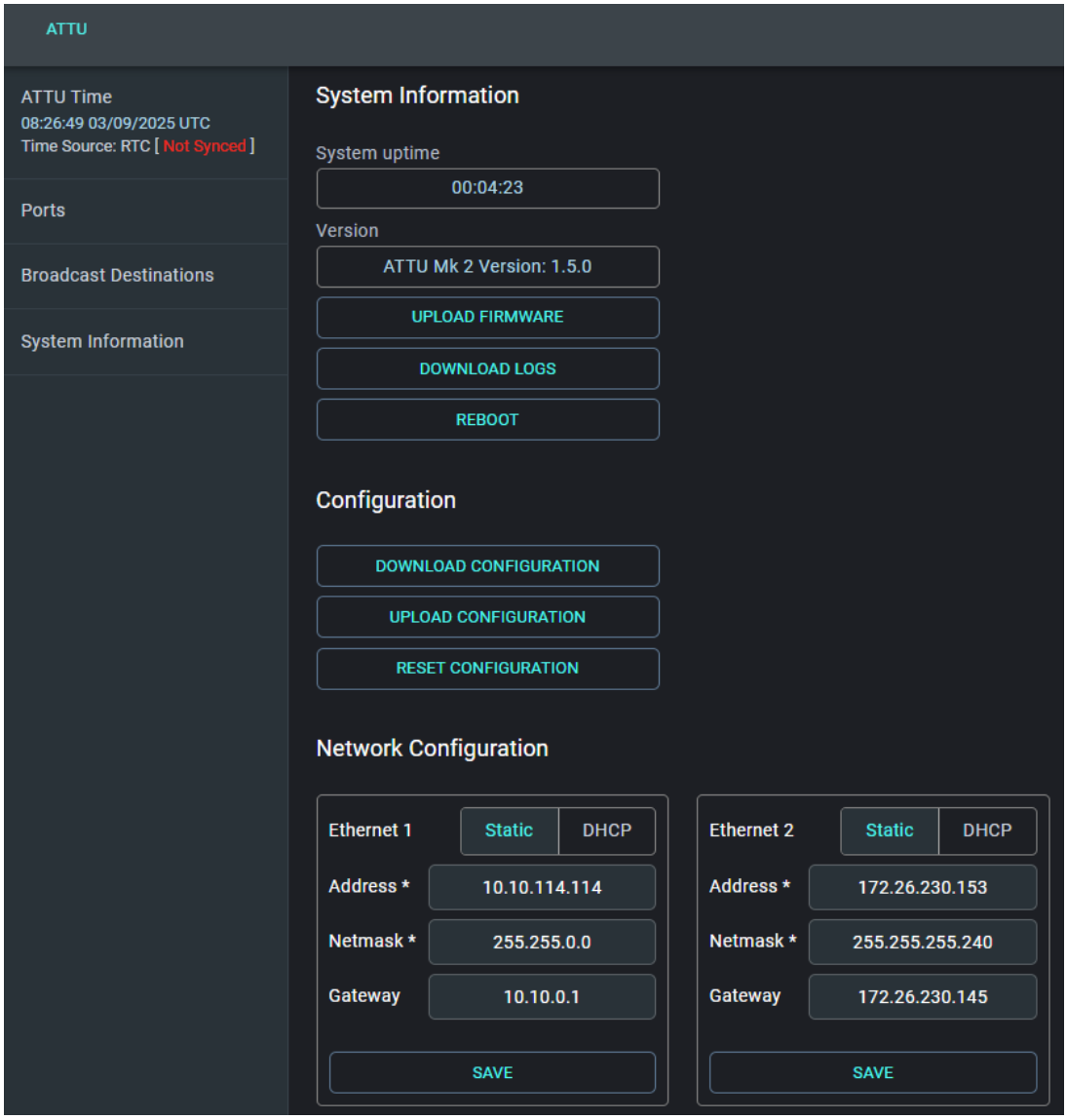


Figure 23 System information

The options on this page are described below:

- **UPLOAD FIRMWARE** – Allows for the upgrade of the ATTU firmware when it becomes available. A .BIN file is uploaded from the monitoring computer to the ATTU.
- **DOWNLOAD LOGS** – Are service log files that are stored in the ATTU. The files should be sent to EIVA upon request.
- **REBOOT** – Performs a soft reboot of the ATTU and is used to clear potential error conditions.
- **DOWNLOAD**, **UPLOAD** and **RESET CONFIGURATION** – Allow for saving the configuration to a file and reloading the configuration. Resetting the configuration will return the configuration to factory default and will reset all incoming ports and broadcasting destinations. This does not count for ATTU IP addresses.

6 Integration

6.1 Connecting to NaviPac

The ATTU Mk II can connect seamlessly to NaviPac. The unit appends timing information to the incoming sensor data to provide the most accurate timing information available.

Once the IP addresses and ports of the NaviPac and NaviScan computers are set up in the **BROADCAST DESTINATION** page, the instruments can be added to NaviPac with the appropriate settings of the ATTU message.

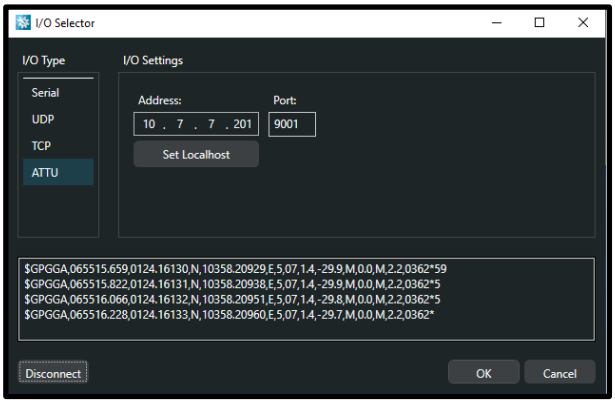


Figure 24 Incoming data

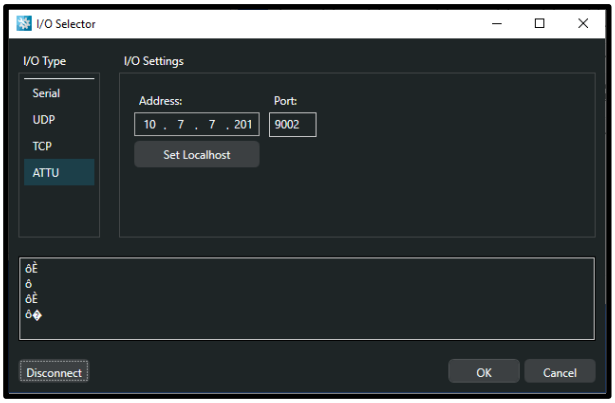


Figure 25 Binary data

The data is shown as binary data from the EM3000 and is not intended to be readable by a user.

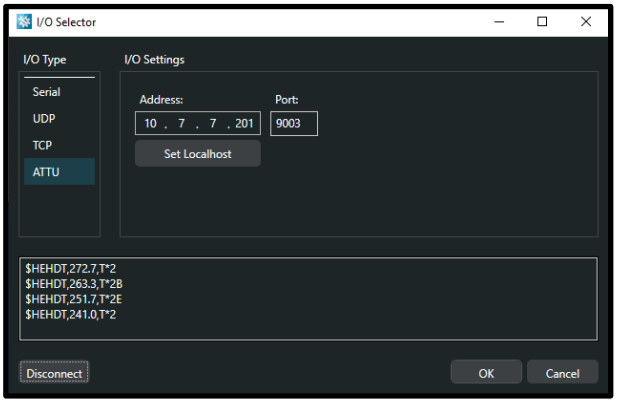
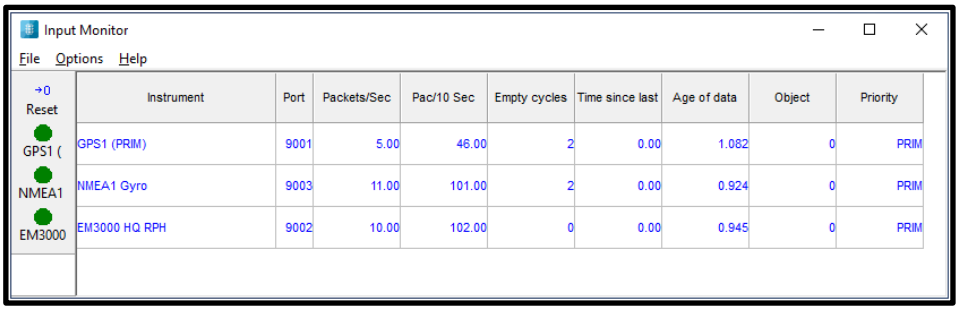


Figure 26 Incoming heading data

The data **Input Monitor** in NaviPac will show the data being received and the frequency of the data. The data is now available for use in NaviPac.



Instrument	Port	Packets/Sec	Pac/10 Sec	Empty cycles	Time since last	Age of data	Object	Priority
GPS1 (PRIM)	9001	5.00	46.00	2	0.00	1.082	0	PRIM
NMEA1 Gyro	9003	11.00	101.00	2	0.00	0.924	0	PRIM
EM3000 HQ RPH	9002	10.00	102.00	0	0.00	0.945	0	PRIM

Figure 27 Input Monitor in NaviPac

6.2 Connecting to NaviScan

The ATTU can connect seamlessly to NaviScan. The ATTU will provide the sensor data with a time-tagged header that is read by the software to improve the timing accuracy of the data.

The setup of the ports is similar to setting up NaviPac and includes the IP address and port number selected for use.

As shown below, the NaviScan computer's IP address is entered with a suitable name, and the ports are added here. Further editing of the settings is available later.

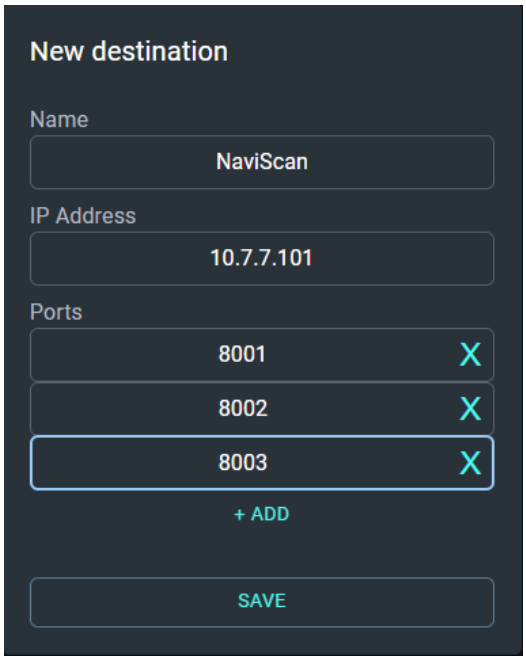


Figure 28 Setting connection to NaviScan

Click **SAVE** to save the settings.

Click the port to select the correct message to send to the NaviScan computer.

Figure 29 Destination of NaviScan computer with allotted ports

Set up in NaviScan involves adding the instruments to the NaviScan Configuration file:

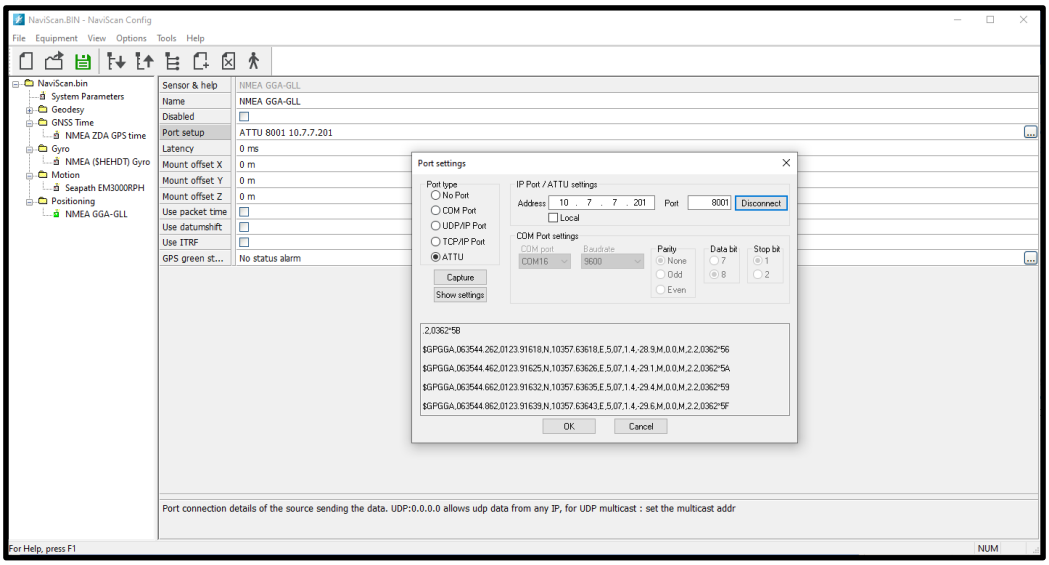


Figure 30 Adding instruments to the NaviScan Configuration

Connect will allow you to see the live data coming into the NaviScan computer.

Once the data is being received and the BIN file is saved, NaviScan Online can be started, and will show the instruments in the Config file.

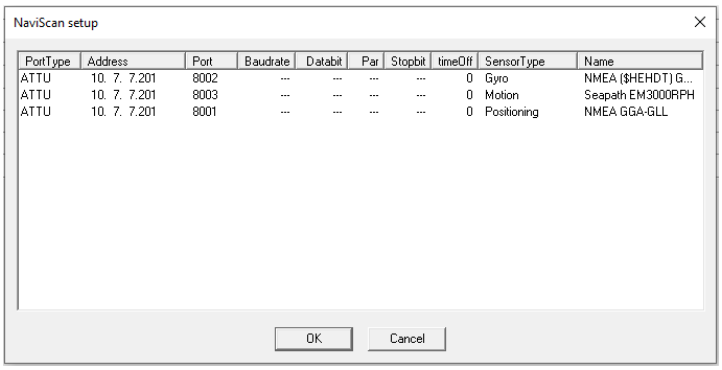


Figure 31 NaviScan Configuration starting Online

When NaviScan Online opens, the instruments will have a green icon for good data. Any data that is incorrect or not received in the correct time will show as a red icon and raise an alarm.

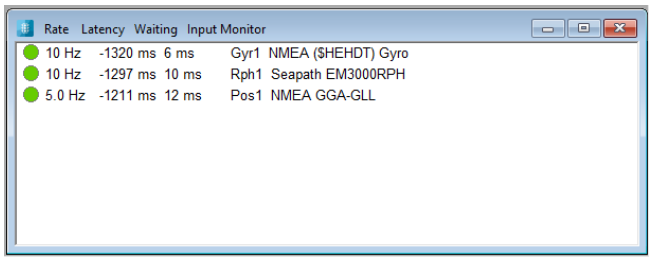


Figure 32 Instrument Monitor in NaviScan Online with good data

To see the incoming raw data, right-click on the **Input Monitor** view to open the **Alarm Settings** and check the **Show Data** checkbox.

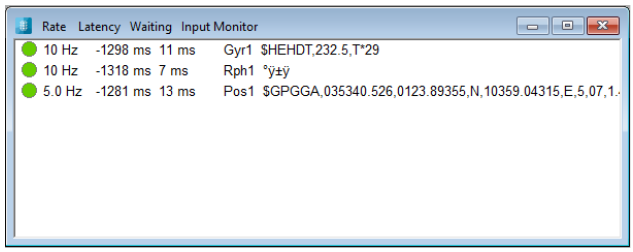


Figure 33 Data shown in Instrument monitor in NaviScan

Other options allow you to see the packet size and data rate in bytes per second. The options to adjust the alarms for the data input are present here too.

6.3 Combined NaviScan and NaviPac

If NaviScan and NaviPac are operated simultaneously, several settings will enable NaviPac to forward incoming data to NaviScan, therefore reducing the number of required output ports from the ATTU.

Firstly, NaviPac requires that the setting for port sharing is checked.

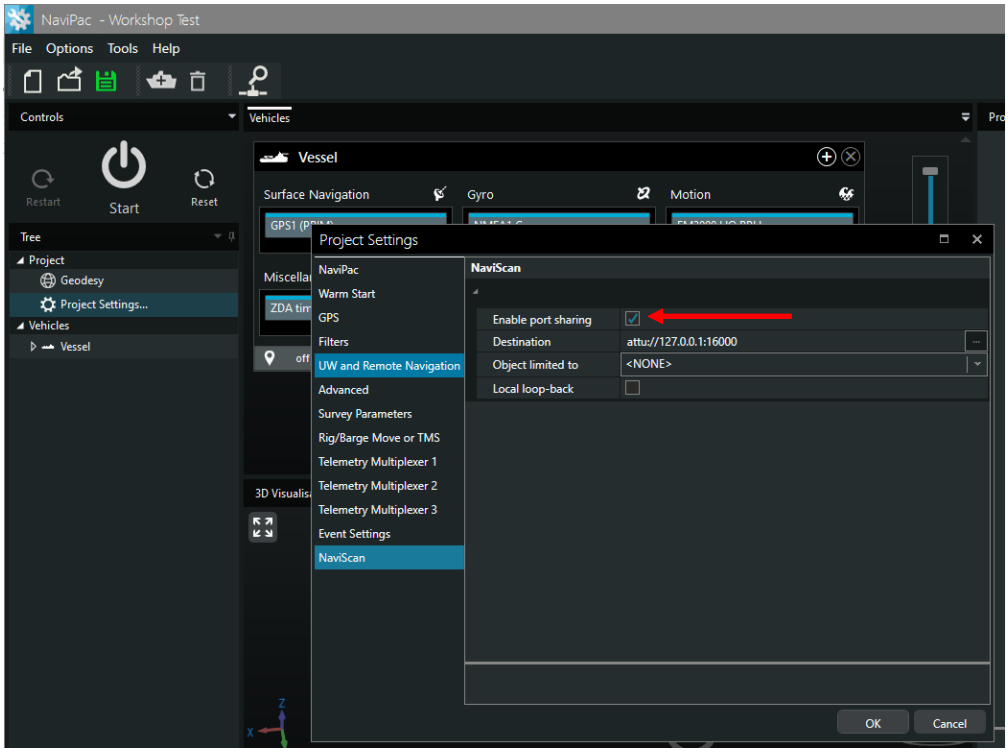


Figure 34 Port sharing from NaviPac to NaviScan

Next, the **Destination** port needs to be noted. In this case, *port 16000* will be added to the incoming port number in NaviPac. By adding 16000 to the input port number (in this case 8001), NaviPac will share the data to NaviScan on port 24001 (8001+16000 = 24001). In the example above NaviScan is running on the same computer as NaviPac, hence the IP address *127.0.0.1*.

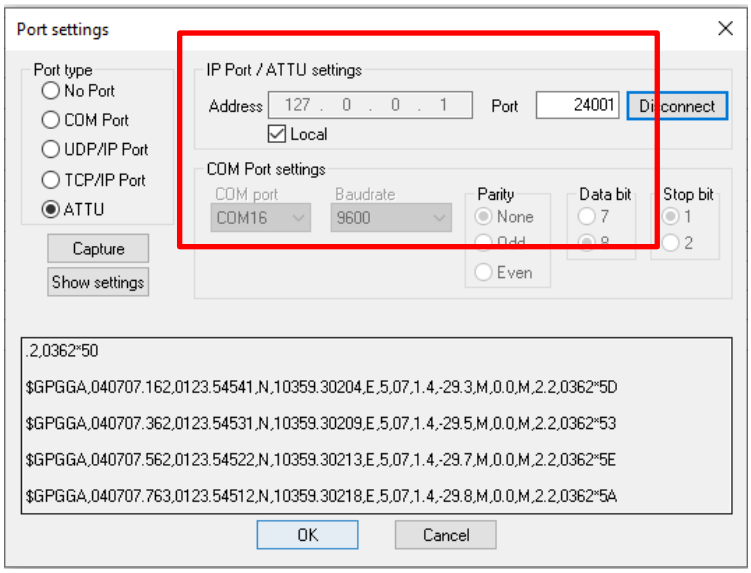


Figure 35 NaviScan port settings with the IP address with the port number 24001

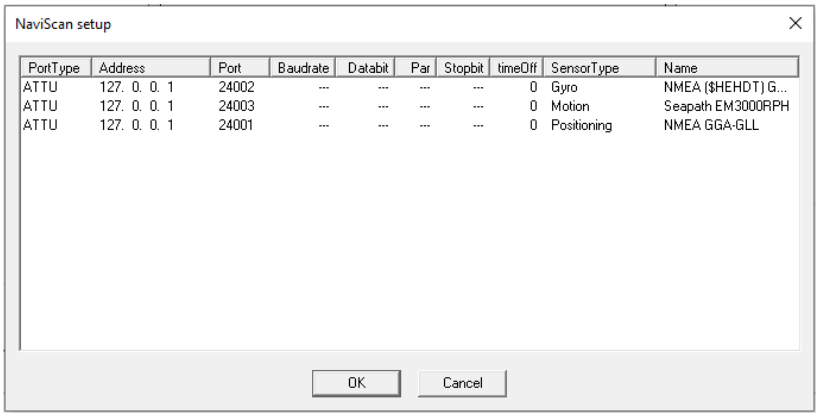


Figure 36 NaviScan incoming equipment settings.

7 Troubleshooting

In case of issues that cannot be resolved by the user of equipment please contact EIVA support on support@eiva.com.

7.1 Occasional yellow flags are showing on the incoming port

This can mean that the port buffer fills up if the data is updated quickly. Increase the baud rate to a higher rate to allow the data to flow faster.

7.2 Data doesn't fit into the entire box in the incoming port

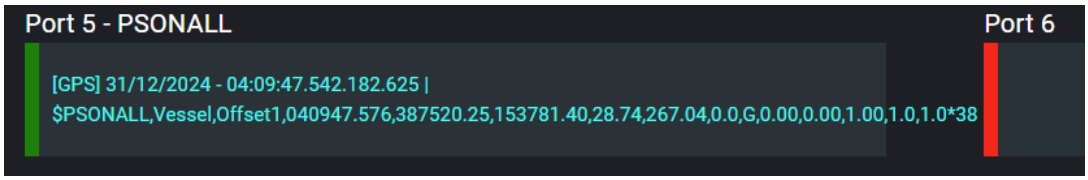


Figure 37 Data shown outside of the bounding box

This happens due to the display scale. Setting the display scale to a lower number should decrease the size of the lettering and allow the screen to fit the entire ATTU web page.

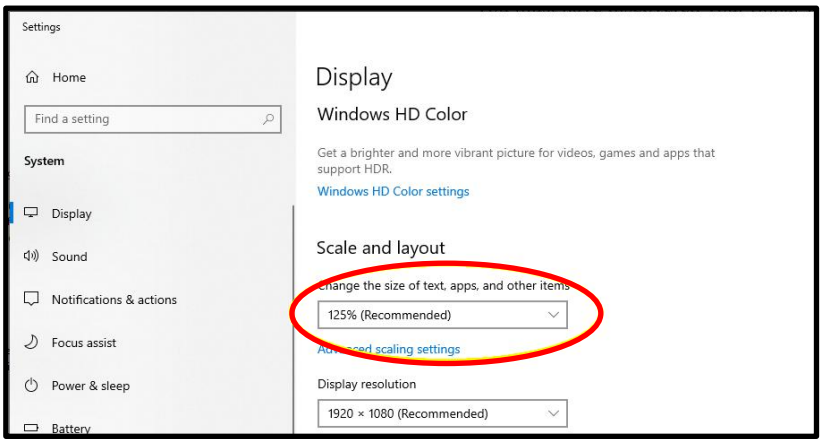


Figure 38 Windows Display scale

7.3 Unable to connect to ATTU

Sometimes, the Linux engine may not start fully and will stay in a suspended state indefinitely. The easiest way to detect this is to start a CMD session and type:

ping IP ADDRESS -t

which will continuously ping the address until stopped.

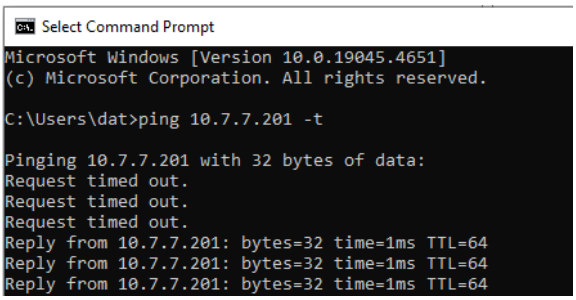


Figure 39 Pinging IP continuously

If the ATTU is suspended, the message will be Request timed out. After 3 messages, cycle the power to the ATTU and wait 5 to 10 seconds.

Repeat cycling the power until the *Reply from* message is received, indicating the ATTU has started correctly and is fully operational.

7.4 Time telegram not being received and interpreted

The ATTU needs to receive ZDA with PPS or UTC with PPS only. The ATTU will disregard the ZDA or UTC when received as a combination of strings such as ZDA and GGA (or any other NMEA strings) as processing the unwanted string can affect the timing.

In contrast to connecting to a GPS ZDA and PPS output, Internet NTP Time Servers can be quite difficult to connect to, even in an office setting. Gateways, firewalls and the like can affect the ATTU unit's ability to connect an NTP server.

The ATTU requires the IP address of the NTP server. This can be found through the Command Prompt with the NSLOOKUP command. Use the computer that is connected to the ATTU to ensure that the NTP server is accessible at the location with a *ping* command.

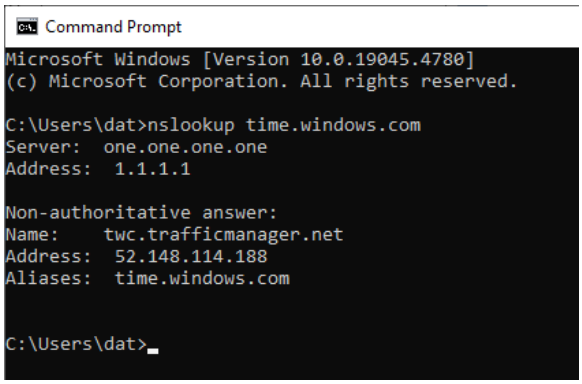


Figure 40 CMD window with commands

7.5 Data not consolidated

The ATTU shows this message in the **Ports** page when the message is not understood, either by having the wrong baud rate, ASCII instead of binary, or wrong port protocol.

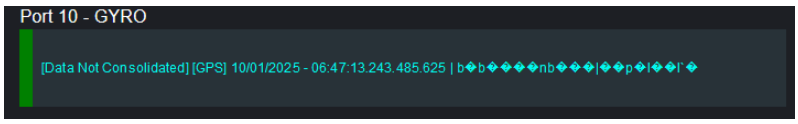


Figure 41 Data not consolidated

7.6 The ATTU does not respond

If the sensors are sending data at a high rate, the ATTU port buffers will fill up, and changing the port parameters does not appear to affect the incoming data.

Waiting for the data to clear from the buffer is reasonable for a slow update like 1 Hz. For high data rates of 5 Hz, rebooting the ATTU will empty the buffers quicker and show the correct data.

7.7 The ATTU is not showing all the data

The ATTU displays the data in an Internet browser like Chrome, Edge or Firefox. Due to limitations in presenting data in PHP within an Internet browser, the data shown on the webpage will lag the ATTU processing unit. This can give the false impression that the ATTU is becoming overloaded. Keeping the browser open and in the foreground will allow the data to “stabilise”.